# Java Exception Handling Report

## FileNotFoundException

### What is FileNotFoundException?

**FileNotFoundException** is a checked exception in Java. It occurs when a program tries to open or access a file, but the specified file cannot be found, or the file cannot be opened. In this example, the program tries to open **abc.txt**. If that file is not available in the expected location, the exception is caught and an appropriate message is displayed.

### Objective

To understand and handle **FileNotFoundException** in Java using a try-catch block while attempting to read a file using the Scanner class.

### Task Description

Write a Java program that creates a File object for **abc.txt** and attempts to open it using Scanner. If the file does not exist or cannot be opened, catch FileNotFoundException and display the message **Exception: File not found** instead of allowing the program to terminate with an unhandled exception.

### Step-wise Execution

**1.** Start the program.

**2.** Import File, FileNotFoundException, and Scanner classes.

**3.** Create a File object referring to the file named abc.txt.

**4.** Create a Scanner object using the File object.

**5.** If abc.txt exists and can be opened, the Scanner is created successfully.

**6.** If the file is missing or cannot be opened, Java throws FileNotFoundException.

**7.** The catch block receives the exception and prints “Exception: File not found”.

**8.** End the program.

### Algorithm

1. START

2. Import required Java I/O and utility classes.

3. Enter the try block.

4. Create File object for abc.txt.

5. Attempt to create Scanner using the file.

6. IF the file is unavailable, FileNotFoundException occurs.

7. Catch the exception.

8. Display “Exception: File not found”.

9. END

### UML Diagram

| FileNotFoundDemo | |
|---|---|
| **+ main(args : String[]) : void** | |
| **main()** | Creates File → Creates Scanner |
| **try** | Attempts to open abc.txt |
| **catch** | FileNotFoundException → print message |

Flow: **Start** → **Create File object** → **Open with Scanner** → **File available?** → Yes: Scanner created → End; No: catch FileNotFoundException → Print message → End.

## Program Code

```
import java.io.File;
import java.io.FileNotFoundException;
import java.util.Scanner;

class FileNotFoundDemo {
    public static void main(String[] args) {
        try {
            File file = new File("abc.txt");
            Scanner sc = new Scanner(file);
        } catch (FileNotFoundException e) {
            System.out.println("Exception: File not found");
        }
    }
}
```

## Code Explanation

**`import java.io.File;`** — Imports the File class, which represents a file or directory path.

**`import java.io.FileNotFoundException;`** — Imports the checked exception thrown when a file cannot be found or opened.

**`import java.util.Scanner;`** — Imports Scanner, which can read data from a file.

**`File file = new File("abc.txt");`** — Creates a File object representing abc.txt; it does not itself read the file.

**`Scanner sc = new Scanner(file);`** — Attempts to open the file. If unavailable, FileNotFoundException is thrown.

**`catch (FileNotFoundException e)`** — Catches the exception so the program can handle the missing/unopenable file.

**`System.out.println("Exception: File not found");`** — Displays a clear error message.

## Output Screenshot

Command Prompt / Terminal

```
C:\Java> java FileNotFoundDemo

Exception: File not found


C:\Java>
```

## Output Explanation

If **abc.txt** is not present in the expected directory or cannot be opened, the Scanner constructor throws FileNotFoundException. The catch block handles it and the output is **Exception: File not found**.

## Important Note

FileNotFoundException is a **checked exception**, so Java requires it to be handled with try-catch or declared using **throws**. Creating a File object does not guarantee that the file exists; the exception occurs when the program actually attempts to open the file.